Shahriar Hooshmand

AI Engineer – Computer Vision & VLM

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Summary

AI engineer with a B.Sc. in Computer Engineering and shipped computer-vision work across document OCR, face anti-spoofing, and real-time detection. Fine-tuned a vision-language model for Persian document parsing, built a multi-task face-authentication model now published on Hugging Face, and deployed YOLO systems on edge hardware. Strongest in PyTorch, OpenCV, and turning a trained model into a usable demo or API.

Skills

- **Languages:** Python (proficient), C/C++ (proficient), Java (familiar), SQL
- **Deep learning & CV:** PyTorch, TensorFlow/Keras, OpenCV, YOLOv8/YOLOv11, EfficientNet, LoRA / PEFT, ONNX, NumPy, Pandas, Matplotlib, Scikit-learn
- **Document AI:** PaddleOCR-VL, vision-language models, OCR evaluation (CER/WER), synthetic dataset generation with per-element bounding boxes and reading order
- **Deployment & tools:** Git/GitHub, Linux, Jupyter, Flask, SQLite, Raspberry Pi, NCNN
- **Earlier experience (not current focus):** Arduino, ESP32, AVR, VHDL, Android
- **Languages:** Persian (native), English (professional working proficiency)

Experience

- **AI & Automation Intern** - Zilber
  - Jun - Sep 2026 (Tir – Shahrivar 1405) - 3 months
  - Learned and applied GitHub workflows in a team setting.
  - Worked hands-on with LoRA fine-tuning and OCR vision-language models, and contributed to AI projects.
  - Gained practical exposure to basic ERP and automation workflows.

Selected Projects

- **Persian Document OCR with PaddleOCR-VL + LoRA (zi_ocr):** Built a 26.8k-page synthetic Persian official-letters dataset and fine-tuned PaddleOCR-VL-1.6 with LoRA (9.39M trainable of 915M). Held-out test (n=5,855): CER 56.36% → 21.54%, letters-only CER 63.66% → 9.07%.
- **Multi-Task Face Authentication & Deepfake Detection (B.Sc. Thesis):** One EfficientNet-B2 backbone, three heads (deepfake, spoof, temporal consistency). SiW-Mv2 anti-spoofing AUC 0.9856, FaceForensics++ deepfake AUC 0.8854 (demo).
- **Iranian License Plate Recognition:** End-to-end YOLOv8 detection + recognition at 92.8% mAP@50, deployed for real-time inference on Raspberry Pi 5 (demo).
- **Brain Tumor Detection & Segmentation Web App:** Flask app with YOLOv11 real-time MRI segmentation and LLM-generated diagnostic summaries (demo).
- **AI-Powered Robotic Arm:** Arduino arm with real-time YOLO detection of electronic components.
- **Other:** Academic Pamphlet Generator (audio + slides → PDF textbook); English App (Flet, live arXiv abstracts); Android Music Player (Java).

Education

B.Sc. Computer Engineering - Kermanshah University of Technology (KUT), Kermanshah, Iran 2022 - 2026

- GPA 3.67/4.00 (18.19/20)

- Ranked in the top 3 academically among graduated students in the Department of Information Technology.
- Ranked 25th nationally in the 30th National Computer Engineering Student Olympiad (2025).
- Ranked in the top 5.3% among 145,000+ participants in the Iranian National University Entrance Exam (Konkur, 2022).
- Teaching Assistant: Computer Architecture (Fall 2024 & Spring 2025), Algorithm Design (Fall 2025).
- Leadership: Editor-in-Chief, Byte Vision scientific publication (2025–2026); Vice-Secretary, Computer Engineering Scientific Association (Fall 2024 – Summer 2025), co-organizing 20+ workshops and seminars for ~200 students.

Research Interests

- Self-supervised and unsupervised learning in computer vision
- Vision Transformers and attention
- Efficient deep learning and model compression
- Adversarial robustness in vision models
- Federated learning for privacy-preserving computer vision.